results. Of course, the operator we're looking at now is just σ_n, and we already have all the values we need. Let's plug them in. Using Eqs. 3.24 and 3.25, along with our known eigenvalues, we can write

$$\langle \sigma_n \rangle = (+1) \cos^2 \frac{\theta}{2} + (-1) \sin^2 \frac{\theta}{2}$$

or

$$\langle \sigma_n \rangle = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}.$$

If you remember your trigonometry, this gives

$$\langle \sigma_n \rangle = \cos \theta,$$

which agrees perfectly with experiment. Yes! We've done it!

Having come this far, you might want to try your hand on a slightly more general problem. As before, we start with the apparatus $\mathcal{A}$ pointing in the *z* direction. But now, once the spin has been prepared in the *up* state, we can rotate $\mathcal{A}$ to an arbitrary direction in space for the second set of measurements. In this situation, $n_y \neq 0$. Go ahead and try it.

Exercise 3.4: Let $n_z = \cos \theta$, $n_x = \sin \theta \cos \varphi$, and $n_y = \sin \theta \sin \varphi$. Angles θ and φ are defined according to the usual conventions for spherical coordinates (Fig. 3.2). Compute the eigenvalues and eigenvectors for the matrix of Eq. 3.23.